



AGRICULTURE ANALYSIS REPORT

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Overview

This dataset simulates real-world smart farming operations powered by IoT sensors and satellite data. It captures environmental and operational variables that affect crop yield across 500 farms located in regions like India, the USA, and Africa.

This analysis summarizes key insights on crop yield performance.

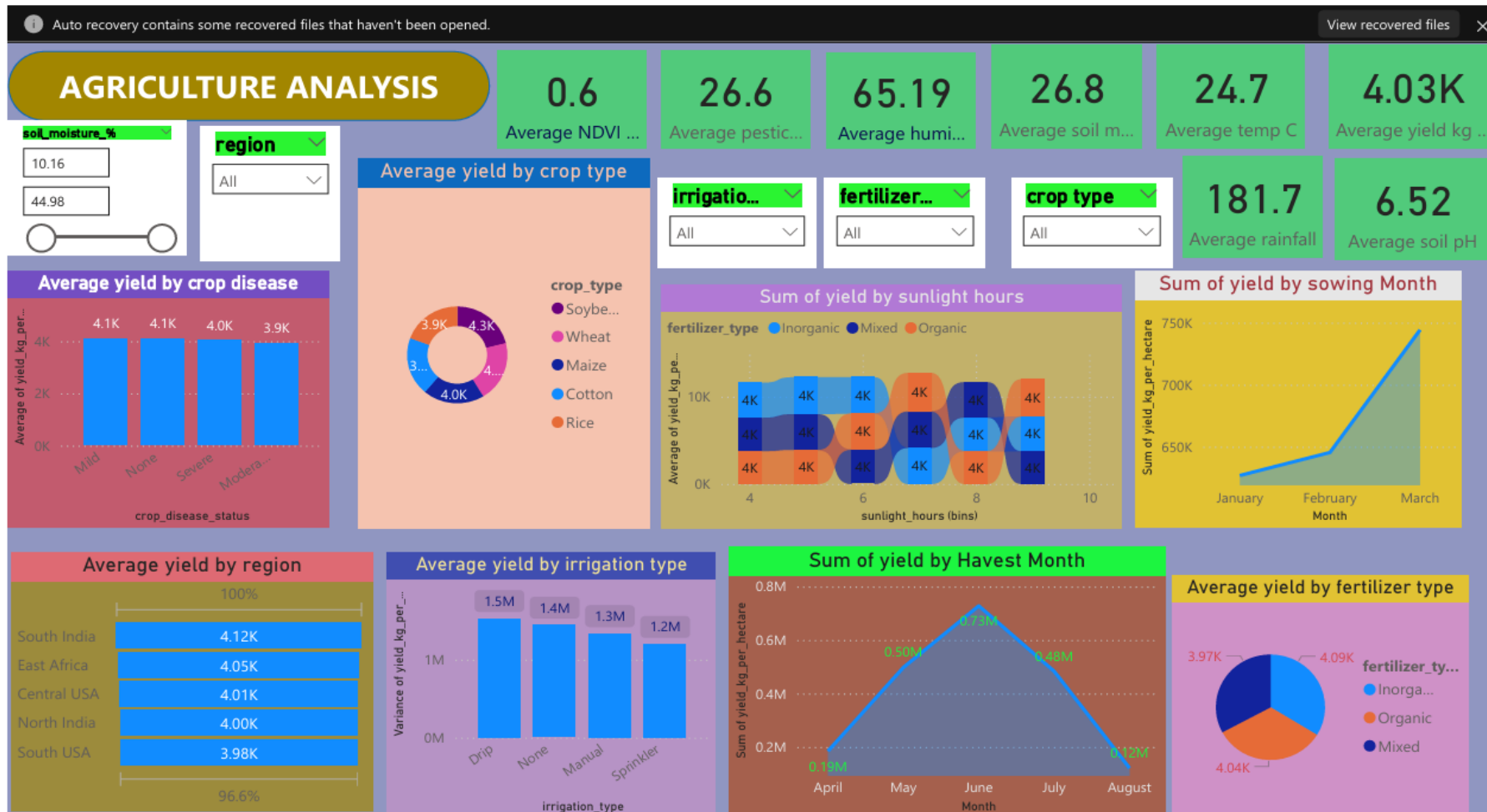
Data cleaning and Transformation

- ▶ Through Power BI, data was visualized using charts like Pie chart, Column chart, Funnel chart Bar chart and Ribbon chart
- ▶ No duplicate was found
- ▶ No missing values found

Questions

- ▶ What is the impact of crop disease, sowing month, crop type, irrigation type, harvest month, fertilizer type, sunlight hours and region crop yield? All of these variables are likely to impact the overall crop yield.
- ▶ Sub questions.
 - ❖ What is the average yield by crop type?
 - ❖ The average yield by crop disease.
 - ❖ What is the average yield by fertilizer type?
 - ❖ The sum of yield by sunlight hour.
 - ❖ The sum of yield by sowing month.
 - ❖ Sum of yield by harvest month.
 - ❖ The average yield by region.
 - ❖ Effect of irrigation type on crop yield.

Data visualization



Data cleaning and Transformation

- ▶ Regional yield differences are small (<150 kg variation), suggesting climate and soil differences are manageable, and good farming practices can close the gap.
- ▶ Inorganic fertilizers produce the highest yields, but organic fertilizer is almost equal—meaning soil health strategies can be effective long-term. Mixed fertilizer is associated with slightly lower performance.
- ▶ Soybean performs the best overall, while cotton and rice show the lowest yields. This may relate to climate sensitivity, nutrient requirements, or irrigation differences.
- ▶ Diseases reduce yield, but the impact becomes more noticeable only at moderate–severe levels. Good crop disease monitoring and early intervention help maintain yields above 4K kg/ha.



THANK YOU